

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1005	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	27.07.2026	Duration:	60 minutes

Code of Conduct

I K NISHALINI (192521301) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K NISHALINI

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

AGILE SOFTWARE ENGINEERING & SYSTEM ARCHITECTURE REPORT

AI-Enabled Smart Forest Fire Detection & Emergency Response System

CHAPTER 1

INTRODUCTION

1.1 Background

Forests are one of the Earth's most valuable natural resources, covering nearly one-third of the world's land area. They play a crucial role in maintaining ecological balance by absorbing carbon dioxide, releasing oxygen, regulating climate, conserving biodiversity, and protecting soil and water resources. Forests are also home to millions of plant and animal species and support the livelihoods of people through timber, medicinal plants, tourism, and agriculture.

Despite their importance, forests are increasingly threatened by wildfires. Forest fires have become more frequent and severe due to global warming, prolonged droughts, rising temperatures, lightning strikes, and human negligence. According to international environmental organizations, millions of hectares of forest land are destroyed every year, resulting in severe environmental, economic, and social consequences.

Traditional methods of forest fire detection mainly rely on human observation, watchtowers, satellite monitoring, or public reporting. These methods are often slow and ineffective because fires are usually detected only after they have grown into large-scale disasters. Delayed detection increases the difficulty of controlling the fire and significantly raises the cost of firefighting operations.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), cloud computing, Geographic Information Systems (GIS), drones, and satellite imaging have enabled the development of intelligent systems capable of monitoring forests continuously. These technologies make it possible to detect fires at an early stage, predict fire spread, and support emergency response teams with accurate information.

The proposed AI-enabled Smart Forest Fire Detection and Emergency Response System combines these modern technologies into an integrated platform capable of continuously monitoring environmental conditions, detecting fire outbreaks automatically, predicting their spread, and providing real-time alerts to emergency authorities.

1.2 Need for the System

The need for an AI-enabled forest fire detection system has become increasingly important due to several environmental and technological challenges.

One of the major challenges in existing forest management is the inability to detect fires during their initial stages. Small fires can spread rapidly because of dry vegetation, strong winds, and high temperatures. By the time they are noticed manually, the fire may have already expanded across large areas.

Human monitoring alone is insufficient because forests often cover thousands of square kilometers, making continuous surveillance impossible. Moreover, remote forest regions have limited accessibility, delaying emergency response.

Artificial Intelligence offers an efficient solution by analyzing real-time sensor data and identifying abnormal environmental conditions within seconds. When integrated with IoT sensors, AI can detect smoke, unusual temperature increases, gas emissions, and humidity changes before flames become visible.

The proposed system reduces dependence on manual observation while providing continuous monitoring, early warning, automated decision-making, and rapid emergency communication.

1.3 Overview of Artificial Intelligence

Artificial Intelligence (AI) refers to the simulation of human intelligence in computer systems that can learn from data, recognize patterns, make decisions, and solve problems automatically.

In forest fire management, AI algorithms analyze environmental data collected from multiple sources, including:

- Temperature sensors
- Humidity sensors
- Smoke sensors
- Carbon monoxide sensors
- Thermal cameras
- Satellite images
- Drone surveillance
- Weather forecasting systems

Machine learning models are trained using historical wildfire data to identify conditions that commonly precede forest fires. Once trained, these models continuously evaluate incoming sensor data and estimate the probability of fire occurrence.

Deep learning techniques, especially Convolutional Neural Networks (CNNs), can analyze images captured by drones or surveillance cameras to detect smoke and flames with high accuracy. These models significantly reduce false alarms and improve detection speed.

1.4 Internet of Things (IoT) in Forest Monitoring

The Internet of Things (IoT) is a network of interconnected devices capable of collecting and exchanging information through the internet.

In the proposed system, IoT devices are installed throughout the forest to monitor environmental conditions continuously.

Each sensor node measures important parameters such as:

- Air temperature
- Relative humidity
- Smoke concentration
- Carbon monoxide level
- Wind speed
- Atmospheric pressure

These sensors transmit data wirelessly to a cloud server using communication technologies such as LoRaWAN, ZigBee, Wi-Fi, or 5G.

The collected information is analyzed by AI models to determine whether the environmental conditions indicate a potential forest fire.

1.5 Importance of Early Fire Detection

Early detection is one of the most critical factors in controlling forest fires.

Research shows that extinguishing a fire during its initial stage requires significantly fewer resources than controlling a large wildfire.

Early detection offers several benefits:

- Prevents large-scale forest destruction.
- Reduces loss of wildlife habitats.
- Minimizes air pollution.
- Protects nearby communities.
- Reduces firefighting costs.
- Improves emergency response time.
- Saves human lives.

The proposed AI-enabled system continuously monitors forests without interruption, allowing emergency teams to respond before fires become uncontrollable.

1.6 Problem Overview

Forest fires are caused by both natural and human-made factors.

Natural Causes

- Lightning strikes
- Volcanic eruptions
- Extremely high temperatures
- Dry vegetation
- Climate change

Human Causes

- Unattended campfires
- Discarded cigarette butts
- Agricultural burning
- Electrical short circuits
- Industrial accidents
- Illegal forest activities

The increasing frequency of these events highlights the need for intelligent monitoring systems capable of operating continuously with minimal human intervention.

1.7 Proposed Solution

The proposed system integrates Artificial Intelligence, IoT, drones, cloud computing, GIS mapping, and emergency communication into a unified smart platform.

The working process includes:

1. Environmental data collection through IoT sensors.
2. Wireless transmission of data to the cloud.
3. AI analysis of environmental conditions.
4. Detection of potential fire outbreaks.
5. GPS identification of affected locations.
6. Fire spread prediction using machine learning.
7. Automatic notification to emergency authorities.
8. Continuous monitoring until the fire is controlled.

This integrated approach provides faster detection, improved accuracy, and better coordination among emergency response teams.

1.8 Benefits of the Proposed System

The proposed AI-enabled Smart Forest Fire Detection and Emergency Response System offers numerous advantages:

- Real-time environmental monitoring.
- AI-based early fire prediction.
- Automatic emergency notifications.
- Accurate GPS location tracking.
- Reduced response time.
- Cloud-based data storage and analytics.
- Improved firefighter safety.
- Protection of wildlife and biodiversity.
- Reduced economic losses.
- Scalable deployment across large forest areas.

CHAPTER 2

PROBLEM STATEMENT, OBJECTIVES AND SCOPE

2.1 Introduction

Forest fires are one of the most devastating natural disasters affecting ecosystems, wildlife, human settlements, and the global environment. Every year, millions of hectares of forests are destroyed due to natural causes such as lightning strikes and prolonged droughts, as well as human activities including unattended campfires, illegal burning, discarded cigarettes, and industrial accidents. The increasing impact of climate change has further intensified the frequency and severity of wildfires, making early detection and rapid response more important than ever.

Traditional fire detection methods depend largely on human observation, watchtowers, satellite imagery, or public reporting. These approaches often fail to detect fires during their early stages because of limited monitoring coverage, communication delays, and environmental challenges. As a result, fires spread rapidly before emergency teams can respond, leading to significant environmental, economic, and social losses.

To overcome these limitations, an intelligent system capable of continuous monitoring, early fire prediction, automatic alert generation, and efficient emergency response is essential. Artificial Intelligence (AI), Internet of Things (IoT), cloud computing, drones, Geographic Information Systems (GIS), and Machine Learning (ML) provide the necessary technologies to develop such a smart forest fire management system.

2.2 Problem Statement

Forest fires continue to cause extensive destruction because most existing detection systems identify fires only after they have become large and difficult to control. Remote forest regions often lack continuous monitoring, resulting in delayed detection and slow emergency response.

The major problems associated with existing forest fire monitoring systems include:

- Lack of continuous real-time monitoring.
- Delayed fire detection due to manual observation.
- High dependence on human surveillance.
- Limited monitoring coverage in remote forest regions.
- Poor coordination between monitoring stations and emergency services.
- Inaccurate prediction of fire spread.
- High operational and maintenance costs.
- Increased damage to forests, wildlife, and nearby communities.

Therefore, there is a need for an intelligent, AI-based monitoring system that continuously analyzes environmental conditions, predicts possible fire outbreaks, identifies the exact fire location, and automatically notifies emergency response teams to minimize damage.

2.3 Existing Problems

The current forest fire detection process faces several practical challenges that reduce its effectiveness.

1. Delayed Fire Detection

Most traditional systems identify forest fires only after smoke or flames become visible. By this stage, the fire may have already spread over a large area.

Impact:

- Increased forest destruction

- Higher firefighting costs
- Greater risk to human life

2. Human Dependency

Forest monitoring heavily depends on forest guards, watchtower operators, and public reporting.

Limitations:

- Human fatigue
- Observation errors
- Limited monitoring hours
- Slow reporting process

3. Limited Monitoring Coverage

Large forests extend over hundreds or thousands of square kilometers, making continuous manual observation impossible.

Remote regions often have:

- No surveillance cameras
- No communication network
- Difficult terrain
- Limited accessibility

4. Environmental Challenges

Heavy rainfall, fog, smoke, clouds, and dense vegetation reduce the effectiveness of satellite monitoring and manual observation.

5. Lack of Predictive Analysis

Existing systems generally detect fires after ignition but cannot predict fire risk before the fire actually starts.

Without prediction:

- Emergency teams cannot prepare in advance.
- Fire spread becomes difficult to control.
- Resources cannot be allocated efficiently.

6. Slow Emergency Communication

Many monitoring systems require manual verification before notifying emergency services.

This increases:

- Response time
- Damage to forests
- Risk to firefighters

2.4 Causes of Forest Fires

Forest fires occur due to both natural and human-made factors.

Natural Causes

- Lightning strikes
- Extremely high temperatures
- Long periods of drought
- Strong winds
- Volcanic eruptions
- Dry vegetation caused by climate change

Human Causes

- Carelessly discarded cigarettes
- Unattended campfires
- Agricultural burning
- Illegal forest activities
- Electrical equipment failure
- Industrial accidents
- Fireworks
- Vehicle sparks

Studies show that human activities are responsible for a significant proportion of forest fires worldwide, emphasizing the importance of continuous monitoring and awareness.

2.5 Effects of Forest Fires

Forest fires have severe consequences on the environment, economy, and society.

Environmental Effects

- Destruction of forest ecosystems
- Loss of biodiversity
- Air pollution
- Soil degradation
- Climate change acceleration
- Carbon dioxide emissions

Economic Effects

- Loss of timber resources
- Damage to agricultural land
- Increased firefighting expenses
- Infrastructure destruction
- Tourism losses

Social Effects

- Loss of human life
- Displacement of communities
- Health problems caused by smoke
- Water pollution
- Wildlife habitat destruction

2.6 Objectives of the Proposed System

The proposed AI-enabled Smart Forest Fire Detection and Emergency Response System aims to achieve the following objectives:

Primary Objectives

- Detect forest fires during the earliest possible stage.
- Monitor forests continuously using IoT sensors.
- Analyze environmental data using Artificial Intelligence.
- Predict possible fire outbreaks using Machine Learning algorithms.
- Automatically identify fire locations using GPS.
- Send instant alerts to emergency authorities.
- Improve response time for firefighting operations.
- Reduce forest destruction and wildlife loss.

Secondary Objectives

- Store historical environmental data.
- Support data visualization through dashboards.
- Improve prediction accuracy over time.
- Reduce dependence on manual monitoring.
- Minimize operational costs.
- Increase public safety.

2.7 Scope of the Project

The proposed system can be implemented in various environments where wildfire monitoring is essential.

Application Areas

- National forests
- Wildlife sanctuaries
- Reserved forests
- Mountain forests
- Agricultural plantations
- National parks
- Eco-tourism zones
- Industrial forest regions

Functional Scope

The system provides:

- Continuous environmental monitoring
- Real-time fire detection
- Smoke analysis
- Temperature monitoring
- Humidity analysis
- GPS location tracking
- AI-based fire prediction
- Emergency notification
- Dashboard visualization
- Historical data storage

- Fire spread prediction

Technical Scope

The system integrates:

- Artificial Intelligence
- Machine Learning
- Internet of Things (IoT)
- Cloud Computing
- GPS Technology
- Drone Surveillance
- Satellite Monitoring
- Geographic Information Systems (GIS)

2.8 Proposed Solution

The proposed solution combines modern technologies to create a fully automated forest fire detection platform.

Step 1 – Data Collection

IoT sensors deployed throughout the forest collect:

- Temperature
- Humidity
- Smoke concentration
- Carbon monoxide level
- Wind speed
- Atmospheric pressure

Step 2 – Data Transmission

Sensor data is securely transmitted to a cloud server using wireless communication technologies such as LoRaWAN, ZigBee, Wi-Fi, or 5G.

Step 3 – AI Analysis

Machine learning algorithms continuously analyze environmental conditions and determine the probability of fire occurrence.

Step 4 – Fire Detection

When abnormal conditions are detected, the system confirms the possibility of fire using AI-based decision models.

Step 5 – Emergency Response

The system immediately:

- Identifies the GPS location.
- Sends alerts to forest officers.
- Notifies nearby fire stations.
- Displays the affected area on a digital map.
- Predicts fire spread direction.
- Recommends the fastest emergency route.

2.9 Expected Outcomes

The proposed system is expected to provide the following benefits:

- Faster fire detection
- Reduced emergency response time
- Improved prediction accuracy
- Better forest conservation
- Enhanced firefighter safety
- Reduced environmental damage
- Efficient disaster management
- Lower operational costs
- Increased public awareness
- Reliable real-time monitoring

CHAPTER 3

SYSTEM ANALYSIS

3.1 Introduction

System analysis is one of the most important phases in software engineering. It involves studying the existing system, identifying its limitations, gathering user requirements, and proposing an efficient solution to solve the identified problems. The purpose of system analysis is to understand how the proposed system can improve the overall performance, reliability, and effectiveness of forest fire monitoring.

In the proposed AI-enabled Smart Forest Fire Detection and Emergency Response System, system analysis focuses on evaluating the shortcomings of traditional fire detection methods and designing an intelligent solution that integrates Artificial Intelligence (AI), Internet of Things (IoT), Machine Learning (ML), Geographic Information Systems (GIS), cloud computing, drones, and emergency communication systems.

The analysis helps determine the technical feasibility, operational feasibility, economic feasibility, and overall effectiveness of the proposed solution.

3.2 Existing System

Traditional forest fire detection systems mainly rely on manual observation and conventional monitoring techniques.

The commonly used fire detection methods include:

- **Watch Towers**

Forest officers monitor forests from elevated towers and report smoke or fire manually.

- **Satellite Monitoring**

Satellites capture forest images periodically to identify fire hotspots.

- **Drone Surveillance**

Drones fly over forests and capture aerial images for fire detection.

- **CCTV Surveillance**

Cameras installed at strategic locations continuously monitor forest regions.

- **Public Reporting**

Citizens or travelers report fire incidents to nearby authorities.

Although these methods are useful, they suffer from several limitations.

3.2.1 Limitations of Existing System

The existing system faces many operational challenges.

1. Delayed Detection

Fire is detected only after smoke or flames become visible.

2. Human Dependency

Monitoring depends on forest officers and public reporting.

3. Limited Coverage

Large forest regions cannot be monitored continuously.

4. High Operational Cost

Continuous human monitoring requires significant manpower and infrastructure.

5. Weather Dependency

Heavy rain, fog, and clouds reduce the effectiveness of surveillance.

6. No Intelligent Prediction

The existing system cannot predict possible fire outbreaks before ignition.

7. Slow Emergency Communication

Emergency response depends on manual verification and reporting.

Problems Identified

- Slow fire detection
- Poor prediction capability
- Limited monitoring coverage
- High maintenance cost
- Human errors
- Lack of automation
- Delayed emergency response
- Inefficient resource utilization

3.3 Proposed System

The proposed AI-enabled Smart Forest Fire Detection and Emergency Response System provides an intelligent solution by integrating multiple advanced technologies into a single platform.

The system continuously collects environmental data from IoT sensors installed throughout the forest. The collected data is transmitted to a cloud server where AI algorithms analyze environmental conditions in real time.

Whenever abnormal values such as rising temperature, increasing smoke concentration, or carbon monoxide levels exceed predefined thresholds, the AI model predicts the possibility of a forest fire.

The system immediately sends emergency alerts to firefighters and forest officers while displaying the fire location on a digital map.

The platform also predicts the direction of fire spread based on weather conditions such as wind speed and humidity.

3.4 Features of Proposed System

The proposed system offers several intelligent features.

Continuous Monitoring

Environmental parameters are monitored 24 hours a day.

AI-Based Fire Prediction

Machine Learning algorithms analyze historical and live sensor data to predict fire risk

Real-Time Detection

Smoke and temperature changes are detected instantly.

GPS Tracking

The exact fire location is identified using GPS coordinates.

Emergency Notification

Automatic SMS, email, and mobile app notifications are sent to emergency teams.

Fire Spread Prediction

AI estimates the future direction and speed of fire spread.

Cloud Data Storage

All environmental data are stored securely for future analysis.

Dashboard Visualization

Authorities can monitor the entire forest through an interactive dashboard.

3.5 Comparison Between Existing and Proposed System

Feature	Existing System	Proposed AI System
 Monitoring	 Manual	 Fully Automated
 Fire Detection	 Delayed	 Real-Time
 Prediction	 Not Available	 AI-Based Prediction
 Monitoring Coverage	 Limited	 Large Scale
 GPS Location	 Manual	 Automatic
 Alerts	 Manual	 Instant Notification
 Fire Spread Prediction	 Not Available	 Available
 Data Storage	 Limited	 Cloud Database
 Decision Making	 Human	 Artificial Intelligence
 Response Time	 Slow	 Very Fast

3.6 Feasibility Study

A feasibility study evaluates whether the proposed project can be successfully implemented.

3.6.1 Technical Feasibility

The proposed system uses widely available technologies such as:

- Artificial Intelligence
- Machine Learning
- IoT Sensors
- Raspberry Pi
- Arduino
- Cloud Computing
- GPS
- Drone Technology
- Wireless Communication

These technologies are mature, reliable, and suitable for large-scale deployment.

Result: Technically Feasible.

3.6.2 Economic Feasibility

Although the initial installation cost is relatively high, the system significantly reduces long-term operational expenses.

Cost Savings

- Reduced manpower
- Reduced firefighting expenses
- Lower environmental damage
- Reduced property loss
- Lower maintenance costs

Therefore, the proposed system is economically beneficial.

3.6.3 Operational Feasibility

The system is designed to be user-friendly.

Forest officers can easily monitor the system using:

- Mobile Application
- Web Dashboard
- Emergency Alert System

Minimal technical training is required.

3.6.4 Schedule Feasibility

The project can be completed within approximately 6 months.

3.7 Requirement Analysis

System requirements are divided into functional and non-functional requirements.

3.7.1 Functional Requirements

The system should:

- Monitor environmental conditions.
- Collect sensor data.
- Detect smoke.
- Detect temperature changes.
- Predict fire probability.
- Identify GPS location.
- Notify emergency teams.
- Store historical records.
- Display monitoring dashboard.
- Generate reports.

3.7.2 Non-Functional Requirements

The system should provide:

Reliability : Continuous operation without interruption.

Scalability : Support additional sensor nodes.

Security : Encrypted communication and secure cloud storage.

Performance : Real-time processing of environmental data.

Availability : 24×7 monitoring.

Maintainability : Easy software updates and hardware replacement.

3.8 Advantages of Proposed System

The proposed system provides several significant advantages.

- Early forest fire detection
- Continuous environmental monitoring
- AI-based intelligent decision making
- Automatic emergency alerts
- GPS-based location tracking
- Fire spread prediction
- Reduced response time
- Lower operational cost
- Cloud-based monitoring
- Improved disaster management
- Better wildlife protection
- Reduced environmental pollution

3.9 Challenges

Although the proposed system offers many benefits, several challenges need to be considered.

- Initial deployment cost
- Sensor maintenance in remote forests
- Network connectivity issues
- Weather interference

- Battery replacement for sensor nodes
- AI model retraining
- Data security concerns

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design is a critical phase in the Software Development Life Cycle (SDLC), where the requirements gathered during system analysis are transformed into a structured blueprint for implementation. It defines the architecture, components, modules, interfaces, database structure, and data flow of the proposed system.

The AI-enabled Smart Forest Fire Detection and Emergency Response System is designed as an intelligent, real-time monitoring platform that integrates Artificial Intelligence (AI), Internet of Things (IoT), Cloud Computing, Geographic Information System (GIS), GPS technology, and Wireless Sensor Networks (WSN). The system continuously monitors environmental conditions, predicts fire occurrence using AI models, and automatically alerts emergency services for rapid response.

The design emphasizes scalability, reliability, accuracy, security, and ease of maintenance. Each component of the system has a specific role, enabling seamless communication between sensors, cloud services, AI models, emergency authorities, and users.

4.2 System Architecture

The proposed system consists of six major layers:

1. Data Collection Layer

Environmental data is collected using IoT sensor nodes deployed throughout the forest.

Sensors include:

- Temperature Sensor (DHT22)
- Humidity Sensor
- Smoke Sensor (MQ-2)
- Carbon Monoxide Sensor (MQ-7)
- Flame Sensor
- Wind Speed Sensor
- GPS Module

2. Communication Layer

The collected data is transmitted using wireless communication technologies such as:

- LoRaWAN
- ZigBee
- Wi-Fi
- GSM/4G/5G

The communication layer ensures secure and reliable transmission of sensor data to the cloud server.

3. Cloud Processing Layer

The cloud server performs:

- Data storage

- Data preprocessing
- AI model execution
- Fire prediction
- Historical data management
- Dashboard updates

Cloud platforms such as Firebase, AWS IoT, or Microsoft Azure IoT Hub can be used.

4. Artificial Intelligence Layer

The AI engine processes incoming data using machine learning algorithms.

Functions include:

- Fire risk prediction
- Smoke pattern recognition
- Image classification
- Fire spread estimation
- False alarm reduction

Popular AI algorithms include:

- Random Forest
- Decision Tree
- CNN (Convolutional Neural Network)
- LSTM
- XGBoost

5. Emergency Response Layer

When fire is detected:

- GPS location is identified.
- Fire spread direction is estimated.
- Emergency alerts are generated.
- Fire station is notified.
- Forest officers receive SMS/App notifications.
- Disaster Management Center receives alerts.

6. User Interface Layer

Users access the system through:

- Web Dashboard
- Mobile Application
- GIS Map
- Emergency Alert Panel
- Report Generation Module

4.3 Overall System Architecture Diagram

4.4 System Workflow

The proposed system follows the workflow shown below.

4.5Module Design

The proposed system consists of seven functional modules.

Module 1 – Sensor Monitoring Module

Purpose

Collect environmental data.

Inputs

- Temperature
- Humidity
- Smoke
- CO Gas
- Wind Speed

Outputs

Real-time environmental data.

Module 2 – Communication Module

Purpose

Transmit sensor readings to the cloud.

Functions:

- Wireless communication
- Data encryption
- Error checking

Module 3 – AI Prediction Module

Purpose

Predict wildfire occurrence.

Functions:

- Data preprocessing
- Feature extraction
- Machine learning prediction
- Fire classification

Module 4 – GPS Tracking Module

Identifies:

- Latitude
- Longitude

- Forest region
- Fire zone

Module 5 – Emergency Notification Module

Automatically sends:

- SMS
- Mobile notifications
- Email alerts
- Fire station alerts

Module 6 – Dashboard Module

Displays:

- Fire location
- Sensor values
- Live status
- Historical reports
- Fire spread prediction

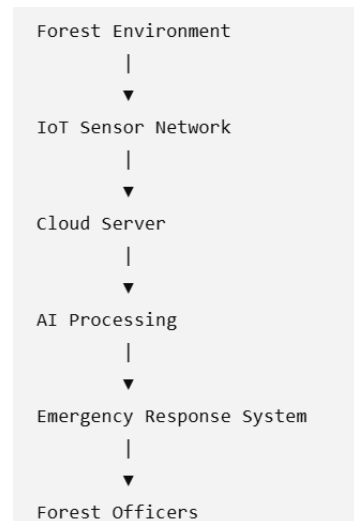
Module 7 – Database Module

Stores:

- Sensor readings
- Fire history
- User information
- Emergency logs

4.6 Data Flow Diagram (DFD)

Level 0 DFD



Level 1 DFD

```
Sensors
|
Collect Data
|
Cloud Database
|
AI Analysis
|
Fire Prediction
|
GPS Tracking
|
Alert Generation
|
Dashboard
|
Emergency Team
```

4.7 Database Design

The database stores all system information

4.8 Input Design

The system receives inputs from multiple sources.

Sensor Inputs

- Temperature
- Smoke
- Humidity
- CO Gas
- Wind Speed

User Inputs

- Login
- Emergency confirmation
- Report generation

4.9 Output Design

The outputs generated by the system include:

- Fire detected notification
- GPS coordinates
- Dashboard visualization
- SMS alerts
- Fire spread prediction map
- Historical reports
- Sensor analytics
- Emergency response status

4.10 Security Design

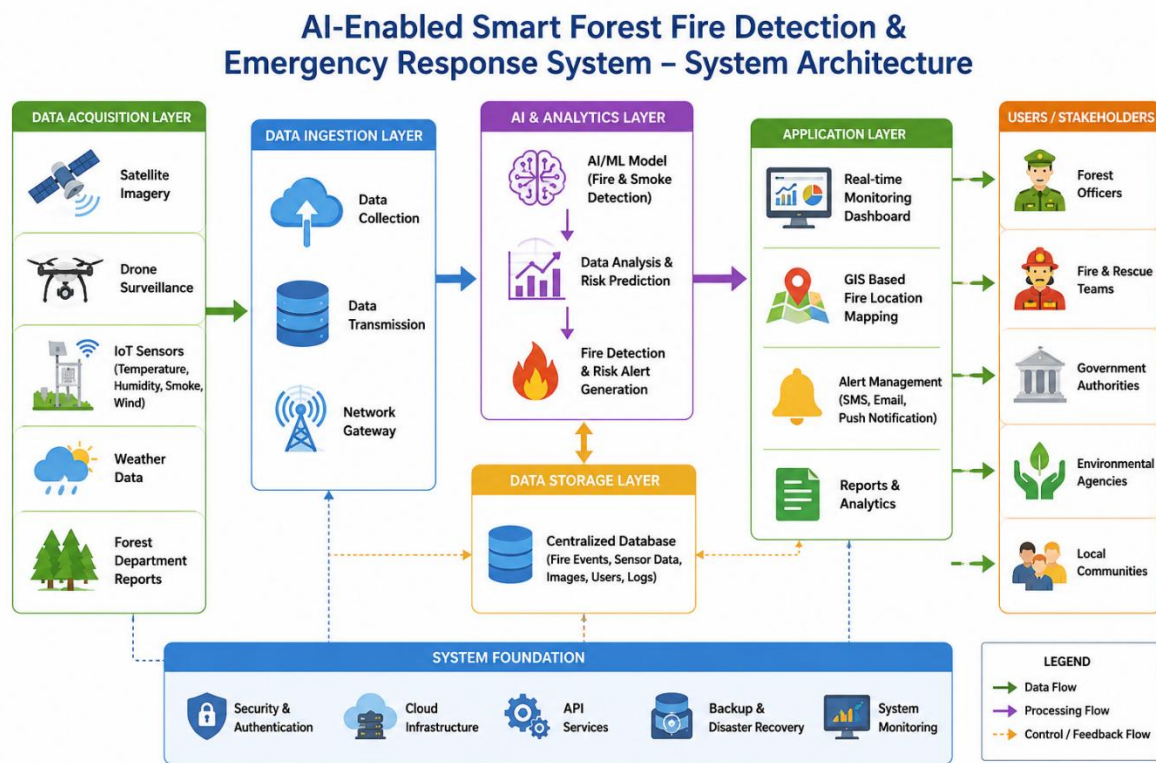
The system implements:

- User authentication
- Password encryption
- Secure cloud storage
- Data encryption during transmission
- Role-based access control
- Backup and recovery mechanism

4.11 Design Advantages

The proposed design provides several advantages:

- Highly scalable architecture
- Real-time monitoring
- AI-powered prediction
- Reduced false alarms
- Fast emergency response
- Secure cloud integration
- Modular implementation
- Easy maintenance
- High reliability
- Better disaster management



CHAPTER 5

AGILE SPRINT PLANNING AND USER STORY WORKSHOP

5.1 Introduction

Agile Software Development is a modern software engineering methodology that emphasizes flexibility, collaboration, continuous improvement, and iterative development. Unlike traditional software development models, Agile divides the project into small and manageable iterations called Sprints. Each sprint delivers a working part of the software, allowing stakeholders to review progress and provide feedback before the next sprint begins.

For the AI-enabled Smart Forest Fire Detection and Emergency Response System, Agile methodology is well suited because the project integrates multiple technologies such as Artificial Intelligence (AI), Internet of Things (IoT), Cloud Computing, GPS, Geographic Information Systems (GIS), and Emergency Notification Systems. Developing these features incrementally ensures better quality, easier testing, and continuous improvements.

The Agile Sprint Planning process helps the development team prioritize user requirements, estimate effort, define sprint goals, and deliver valuable software features within a fixed time frame.

5.2 Agile Methodology

The proposed project follows the Scrum Framework, one of the most widely used Agile methodologies.

Agile Life Cycle

5.3 Project Vision

Vision Statement

"To develop an intelligent AI-enabled Smart Forest Fire Detection and Emergency Response System that continuously monitors forest environments, detects fires at an early stage, predicts fire spread using Artificial Intelligence, and automatically alerts emergency authorities to minimize environmental damage, protect wildlife, and save human lives."

5.4 Project Objectives

The project aims to achieve the following objectives:

- Detect forest fires at the earliest possible stage.
- Monitor environmental conditions continuously.
- Reduce false fire alarms.
- Improve emergency response time.
- Provide accurate GPS-based fire location.
- Predict wildfire spread using AI.
- Reduce forest destruction.
- Improve firefighter safety.
- Store historical environmental data.
- Generate analytical reports.

5.5 Stakeholder Identification

Stakeholders are individuals or organizations directly or indirectly involved in the project.

5.8 User Stories

User stories describe system requirements from the user's perspective.

Epic E1 – User Login

User Story 1

As a Forest Officer,

I want secure login access,

So that I can monitor the forest safely.

Epic E2 – Sensor Monitoring

User Story 2

As a Forest Officer,

I want to view live sensor readings,

So that I can monitor environmental conditions.

Epic E3 – AI Prediction

User Story 3

As a Disaster Management Officer,

I want AI to predict forest fires,

So that emergency actions can begin immediately

Epic E4 – GPS Tracking

User Story 4

As a Firefighter,

I want the exact fire location,

So that I can reach the affected area quickly.

Epic E5 – Emergency Notification

User Story 5

As a Forest Officer,

I want automatic SMS alerts,

So that response time is reduced.

Epic E6 – Dashboard

User Story 6

As a System Administrator,

I want a dashboard,

So that I can monitor all forest regions.

Epic E7 – Fire Spread Prediction

User Story 7

As a Emergency Officer,

I want fire spread prediction,

So that evacuation planning becomes easier.

Epic E8 – Reports

User Story 8

As a Administrator,

I want fire history reports,

So that future planning becomes easier.

5.9 Acceptance Criteria

Acceptance criteria define when a user story is considered complete.

User Story 1

Given a registered user,

When valid credentials are entered,

Then the system should allow login.

User Story 2

Given sensor nodes are active,

When new environmental data is received,

Then the dashboard should update within 5 seconds.

User Story 3

Given abnormal sensor values,

When AI analyzes the data,

Then the system should classify fire risk accurately.

User Story 4

Given fire detection,

When GPS identifies coordinates,

Then the location should appear on the map.

User Story 5

Given confirmed fire detection,

When alerts are generated,

Then emergency teams should receive SMS, email, and app notifications immediately.

5.10 Product Backlog

5.11 Sprint Planning

The project is divided into four sprints, each lasting **two weeks**.

Sprint 1

- User Authentication
- Database Setup
- Sensor Integration

Sprint 2

- AI Model Development
- Dashboard Design
- GPS Integration

Sprint 3

- Emergency Alert System
- Fire Prediction
- Cloud Integration

Sprint 4

- Testing
- Bug Fixing
- Deployment
- Documentation

5.12 Sprint Backlog

5.14 Sprint Goal

Sprint 1 Goal: Establish the project foundation by implementing user authentication, setting up the database, and integrating IoT sensors.

Sprint 2 Goal: Develop the AI prediction engine, dashboard, and GPS tracking functionality.

Sprint 3 Goal: Enable automated emergency notifications, fire spread prediction, and cloud integration.

Sprint 4 Goal: Complete testing, fix defects, optimize performance, and prepare the system for deployment.

5.15 Expected Sprint Deliverables

At the end of the project, the following deliverables will be available:

- AI-enabled forest fire detection software
- IoT sensor integration
- Fire prediction model
- GPS-based location tracking
- Emergency notification module
- Interactive web dashboard
- Fire history and analytics reports
- Cloud database
- System documentation
- User manual

5.16 Advantages of Agile in this Project

- Faster software development.
- Continuous stakeholder feedback.
- Easy adaptation to changing requirements.
- Incremental delivery of features.
- Better risk management.
- Improved software quality through continuous testing.
- Higher customer satisfaction.
- Better collaboration among AI, IoT, and software teams.
- Early identification of defects.
- Efficient project management.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25